

Safe Agents Roadmapping Workshop



Dr. Mark Maybury, VP Commercialization, mark.t.maybury@lmco.com
IJCAI-ECAI
Bremen, Germany

16 August 2026



Public Information Release Authorization (PIRA) CHQ2026060483. Includes previous Public Information Release Authorization (PIRA) including CHQ2026020430, CHQ2026010413, CHQ2026030448, CHQ2026010407, CHQ2025120405, CHQ2024070194, CHQ2025010252, CHQ2025010253

© 2026 Lockheed Martin Corporation. All Rights Reserved



Safe Agents Roadmap Workshop



WELCOME



IJCAI ECAI
BREMEN 2026

Thank you!



- IJCAI-ECAI
- Authors and Participants
- Workshop Committee



Dr. [Mark Maybury](#), VP Commercialization, Lockheed Martin; Board ACSC
mark.t.maybury@lmco.com (chair)

Prof. [Wolfgang Wahlster](#), Founding Director and CEA, DFKI. Wolfgang.Wahlster@dfki.de

Ms. [Josephine Liu](#), Chair, APAC Tech Policy Committee, Association for Computing Machinery (ACM), yuyinl@acm.org

Dr. [Francesca Rossi](#), IBM Fellow, IBM Global Leader for Responsible AI and AI Governance, T.J. Watson Research Center, Yorktown Heights,
Francesca.Rossi2@ibm.com

Dr. [Mihai Christodorescu](#), Research Scientist, Security and Privacy, Google,
christodorescu@google.com

Future of AI Agents: Promise or Peril?

Nightmare

- Jobless, Poverty
- Privacy Breach
- Physical Harm

AI FOR HUMANITY

- HEALTH
- EDUCATION
- OPPORTUNITY
- SUSTAINABILITY



Motivation

- **Promise and Peril:** Agents rapidly evolving, becoming ubiquitous with increasing dependence, potential value, vulnerabilities, attack surfaces, and consequences
- **Objective:** Collaboratively articulate a roadmap for Agentic AI
- **Method:** Combination of invited speakers and reviewed submissions from commercial, academic and government participants
- **Participants:** Academic researchers and teachers, commercial practitioners, policymakers and regulators, students
- **Interdisciplinary:** Mix of international participants from multiple communities including but not limited to engineering, safety/security, human factors, legal, ethics
- **Multi-domain:** Multiple verticals (e.g., scientific research, customer service, manufacturing, finance, health, defense, ...)

Headlines of Agent and Robot Dangers

Maybury, M. T. 2026. Mitigating Risks in Agents and Robots. IEEE Int Conference on AI and Data Analytics (ICAD). Boston College,, Chestnut Hill, MA. June 2026. PIRA CHQ2025120405.

The Guardian ^{US} Air Canada ordered to pay customer who was misled by airline's chatbot

Tay: Microsoft issues apology over racist chatbot fiasco **BBC**

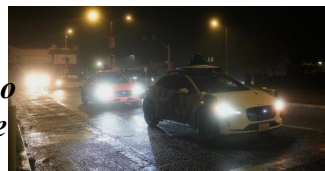
BUSINESS INSIDER 18 December 2023
GM Dealer ChatBot Agrees to Sell 2024 Chevy Tahoe for \$1

US DISTRICT COURT OF MINNESOTA
UnitedHealth "nH Predict" AI denies Medicare-Advantage coverage

MIT Technology Review
A Roomba recorded a woman on the toilet.
How did screenshots end up on Facebook?

FOX NEWS channel
Humanoid robot malfunctions, sparks viral panic
Viral video of out-of-control robot sparks fears, but what really happened?
By Kurt Knutsson, CyberGuy Report - Fox News
Published June 2, 2025 6:00am EDT

The New York Times 21 Dec 2025
Waymo Suspended Service in San Francisco After Its Cars Stalled During Power Outage



AXIOS
Chinese robotics manufacturer left backdoor in product

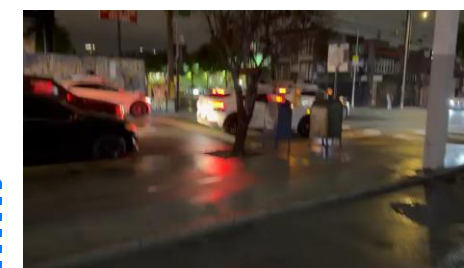
THE ECONOMIC TIMES 27 Feb 2025
Video of robot hitting people in China goes viral, Internet asks "should we be worried?"

KFC apologises after German Kristallnacht promotion **BBC** 10 November 2022

THE WALL STREET JOURNAL. 12 November 2025
Chinese Hackers Used Anthropic's AI to Automate Cyberattacks
The use of AI automation in hacks is a growing trend that gives hackers additional scale and speed

8 CBS8
Tesla faces a lawsuit after a robotic arm knocked a worker unconscious and pinned him to the ground

THE CONVERSATION 27 Feb 2025
A robot nearly headbutted a festival spectator in China



— Court Cases

--- AI Incident DB

..... Journalism

AI Incident DB: Microsoft Tay [6](#), KFC Kristallnacht [410](#), Roomba [521](#), Chevy Tahoe [622](#), Waymo [1326](#),

Related IJCAI-ECAI Workshops

- The 2nd Workshop on [AIGC Safety](#) from Dual Perspectives of Model and Content
- 1st IJCAI Workshop on [Safe Physical AI](#)
- TRUST-AI 2026 – the second European workshop on [Trustworthy AI](#)
- [Ethics in Clinical AI](#) Applications (ETHICAIA)
- Workshop on [Foundation Models for Social Good](#) (FMSG)
- The Fourth Workshop on [Computational Fair Division](#) at IJCAI 2026
- IJCAI 2026 Workshop on [Explainable Artificial Intelligence](#) (XAI)
- EASi-EXPLIMED: A Joint Workshop on [Explainable AI in Space and Explainable Artificial Intelligence for the Medical Domain](#)
- CLEAR Workshop: Collaborative Methods and Tools for Engineering and Evaluating [Transparency in AI](#)
- [Neuro-Symbolic](#) Intelligence for LLMs and Autonomous Agents (NILA)
- Augmented Democracies: [AI for Civic Engagement and Democratic Decision-Making](#)

Some Requirements for Safe and Secure AI

- [EU AI Act](#) (Aug 2026) – AI categorized by risk (prohibited, high, limited, minimal) and mandates for data governance, cybersecurity (adversarial testing, incident reporting), human oversight.
- [OECD AI Principles](#): Inclusive growth, sustainable development, well-being; Rule of law, human rights, democratic values; Transparency and explainability; Robustness, security, and safety: Accountability
- [ISO/IEC 42001](#): Plan-do-check methodology for auditing AI safety
- [US Executive Order 14110](#) (Oct 2023): Safe, Secure and Trustworthy development and use of Artificial Intelligence.
- NIST AI Risk Management Framework ([AI RMF](#))
- [Council of Europe Framework Convention on AI](#) (2024): First legally binding international treaty aimed at ensuring AI aligns with human rights, democracy, and the rule of law.
- General Data Protection Regulation ([GDPR](#)), Data Protection Impact Assessments ([DPIAs](#)), UK [DUAA](#) safeguards for high-risk AI
- Personal Information Protection Law ([PIPL](#), China)



COMMON ELEMENTS

- Risk Management
- Transparency
- Technical Robustness
- Human Oversight
- Data Governance

NIST RFI on Agent Security

- ~1000 responses noting behavioral and systemic threat analyses, emphasizing autonomous agents necessitate a shift from static testing to runtime governance, as they break traditional threat models by altering real-world systems. Key themes include high risk of indirect prompt injections, requirement for "presume breach" isolation of agent environments, and implementation of capability-level access controls to mitigate risks from advanced, dual-use models (Riggs et al 2026). Government roles in (threat) information sharing, standards, and implementation guidance.
- "Agent architectures change core assumptions around code-data separation, authority boundaries, and execution predictability, creating new confidentiality, integrity, and availability failure modes" -> need for layered defense stack (Li et al 2026).
- Lockheed Martin Submission (2025)
 - **National Security:** Importance of access to necessary compute and AI hardware, noting smaller-scale defense companies may not be able to compete with commercial demand.
 - **Classification and Governance:** Need for AI-specific security classification reform to enable "multi-classification training environments" to replicate real-world operational scenarios.
 - **Supply Chain Security:** Frameworks like Cybersecurity Compliance and Risk Assessment (CCRA) for evaluation.

Riggs, J. , Hamin, M. , Perry, N. , Edelman, B. and Cihon, P. (2026), [Summary Analysis of Responses to the Request for Information Regarding Security Considerations for AI Agents](#), NIST Trustworthy and Responsible AI, National Institute of Standards and Technology, Gaithersburg, MD (Accessed June 8, 2026)

Li, N., Zhang, K., Polley, K., & Ma, J. (2026). Security considerations for artificial intelligence agents (Perplexity response to NIST/CAISI request for information 2025-0035). arXiv. arxiv.org/pdf/2603.12230

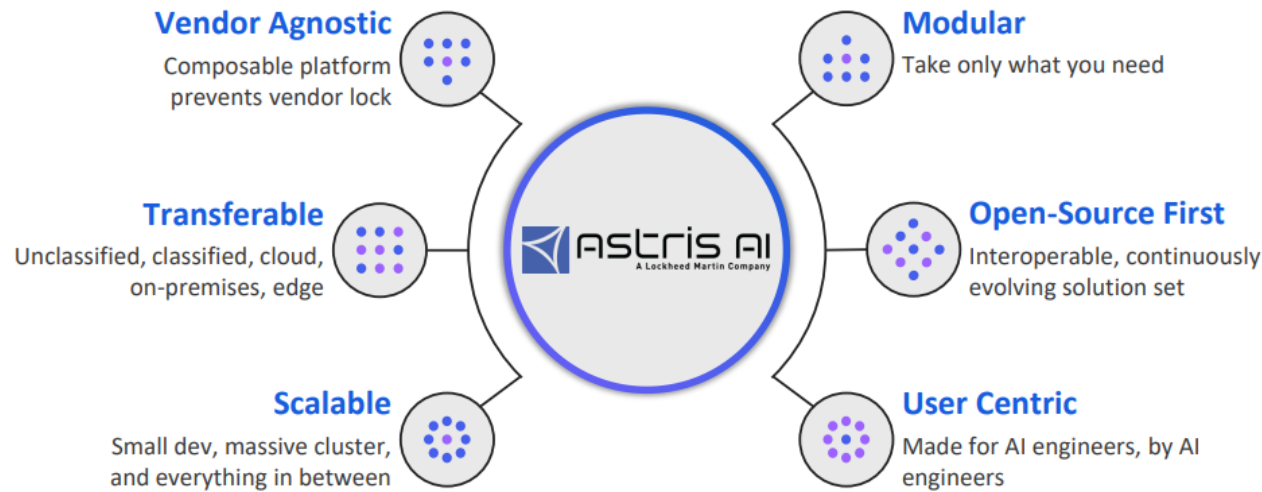
Lockheed Martin Corporation. (2025). Lockheed Martin response to NIST request for information on artificial intelligence (AI) security (Docket No. 250102-0002). Networking and Information Technology Research and Development (NITRD). <https://files.nitrd.gov/90-fr-9088/LMCO-AI-RFI-2025.pdf>



Evolved from Lockheed Martin's Toughest AI Challenges

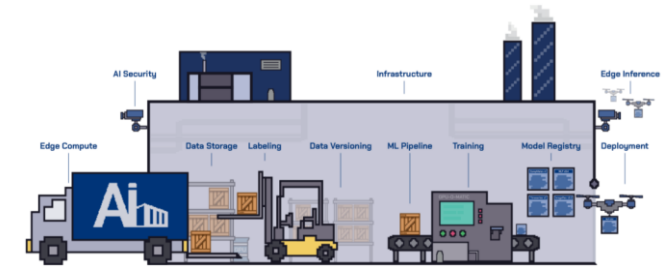
Astris AI originated out of Lockheed Martin's AI Center (LAIC), AI Factory™ for *high-stakes, highly regulated environment*.

Astris AI for secure, compliant, mission-critical enterprise AI deployments



Design Principles

• Open-Source First • Modular • Composable • Scalable • Vendor Agnostic • Transferable • User Centric



Scale demonstrated

80K+
Generative AI users

10k+ AI Engineers and
Developers, **32K** Agents

Processing
4 billion tokens / week

AI vs. Generative AI vs. Agents vs Robots

	Classification/ Prediction
Introduced	80s/90s
Input	Data
Output	Class & Confidence
Technology	AI Algorithm
Control (DOA)	Human in/on loop
Training	Focused data

Increasing Autonomy, Increasing Risk, Increasing Attack Surfaces

AI vs. Generative AI vs. Agents vs Robots

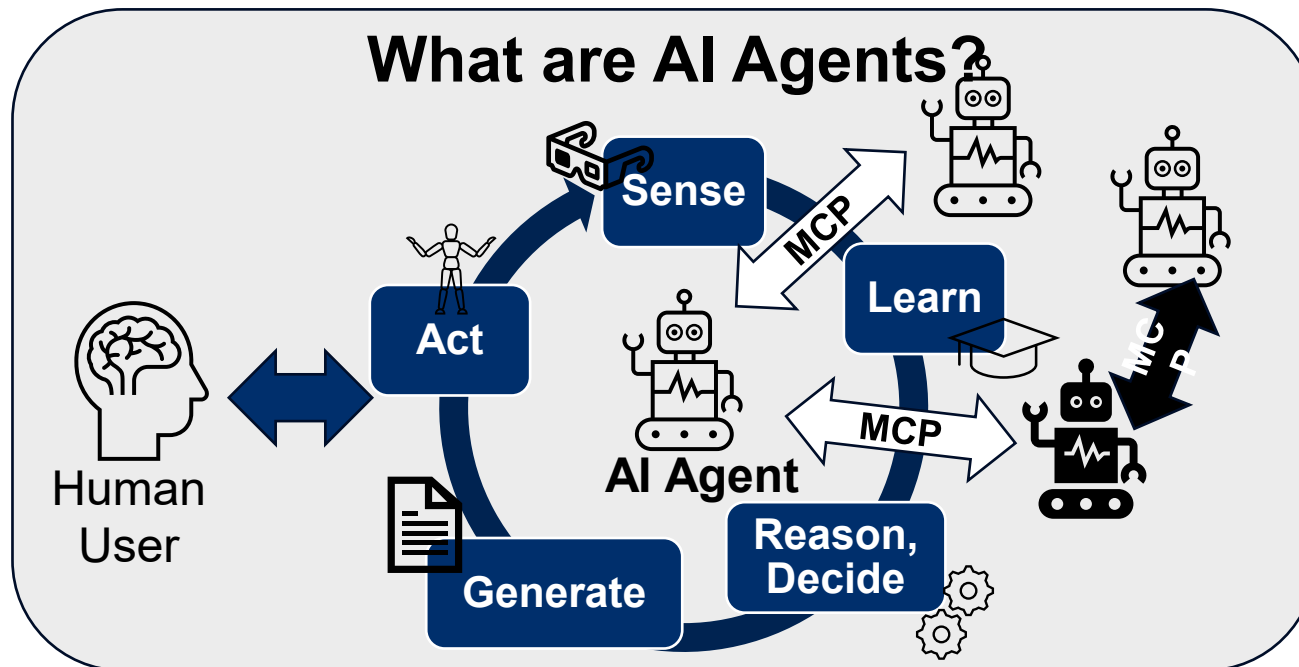
	Classification/ Prediction	Generative AI	Agentic	Robots
Introduced	80s/90s	2022	2025	2026+
Input	Data	Prompt (with context)	Tasking	Tasking
Output	Class & Confidence	Generated text, audio, video, code, molecule	Act or Solve Problem (in digital world)	Physical Action (in real world)
Technology	AI Algorithm	LLMs	LLM + Reasoning	Large world/physics models
Control (DOA)	Human in/on loop	Human in/on loop	Human on or out of loop	Human out of loop
Training	Focused data	Large data	Tasks and solutions	Tasks and (physical) solutions

Increasing Autonomy, Increasing Risk, Increasing Attack Surfaces

Agents

Agentic AI – AI systems that can autonomously perceive (sense), prognosticate (learn), plan/prove/pick (reason/decide) and perform (act) to achieve specific goals within some bounded delegation of authority. See [Google](#) and [AWS](#) definitions.

Model Context Protocol (MCP) - an open-source framework developed by Anthropic to standardize how AI applications, particularly Large Language Models (LLMs), connect to and interact with external data sources and tools. Two-way connection using a JSON-RPC 2.0-based protocol to allow LLMs to access real-time data, perform actions, and utilize specialized features beyond their training data.

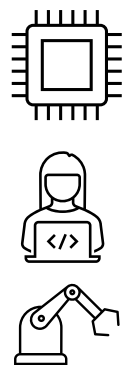


Examples of Agents

- Shopping Assistant 
- Financial Analyst \$
- Legal Analyst 
- Coding Agent 
- Customer service 
- Marketing Agent 
- Supply agent 
- Diagnostic agent 

Safety, Security, and Risk Defined

- **Safety:** Design of technology to prevent accidental or unintentional harm. Harms include Physical, Psychological, Cognitive, Financial, Environmental, Reputational, etc. Safety measures can protection from malfunctions, hazards, alignment with human values, training, governance or policing of proper use, controls, guardrails, etc. [Revised from [ISO/IEC TS 5723:2022](#)]
- **Security:** Avoiding harms by preventing unauthorized use and intentional mis-use of tool or attack on tool by adversary including undermining confidentiality, integrity and availability. [[AI RMF 1.0](#)]
- **Risk** = Threats (to self/others or by others) * Vulnerabilities * Consequences. [NIPP [RMF](#), [DHS Risk Lexicon](#)]



	Safety	Security
Goal	Protect the world from the system	Protect the System from the world
Threat	Unintentional (Errors, Accidents)	Intentional (Malicious)
LLM Example	Misinformation from hallucination or bias	Jailbreaking confidential information with malicious prompts
Agent Example	Revealing personal information without authorization	Manipulating an agent into creating misinformation
Robot Example	Accidentally colliding with a person	Adversary control of a robot

Trustworthy Agents

Safe and Secure

**Bias
Hallucination
Prompt Injection
Guardrails
Alignment
Automated AI BOM
Red Teaming
User Modeling
Modeling Monitoring
Access Control
Risk Management**

**Accurate and
Transparent**

**Robust

Explainable

RAG

Repeatability

Test & Evaluation

Verification and
Validation**

Governed

**AI RMF
Supply Chain
Data Quality
Data Protection
Model Provenance
Benchmarks
Monitoring
Training
Human Oversight**

Workshop Themes



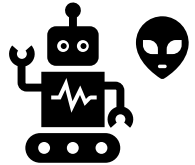
Ethical Foundations and Governance: theory and philosophy of ethical agents

- Assessing current and new ethical frameworks for autonomous and agentic systems.
- The role of law, regulation and policy in guiding agent development.
- Philosophical debate around agent consciousness, rights and responsibilities.



Accuracy, Bias, Fairness, Transparency:

- Methods for improving agent accuracy and self-awareness (e.g., limitations, confidence).
- Successful approaches to detecting and mitigating dangers (e.g., bias, hallucination).
- Techniques for creating more transparent and interpretable models and agents.
- Evaluating the impact of agents on different demographic groups.



Human-Agent Collaboration and Trust: Dynamics of human-agent collaboration:

- Designing intuitive and trustworthy user interfaces for agentic AI.
- Agents in decision support and risks (e.g., awareness, over/under confidence, deskilling).
- Governance frameworks for autonomous decision-making by AI agents.
- Strategies for building public trust and understanding of agent technologies.



Real-world Applications and Case Studies: Responsible Agents in high assurance use cases

- Responsible development and deployment in high assurance applications with human, financial, & security risks.
- Ethical and operational challenges in regulated industries, e.g., healthcare, disaster management, pharmaceutical, financial services, telecom, logistics, energy, agriculture, defense, and space.
- AI in creative fields and impact on human learning, training, and creativity.

Agenda Overview

- Welcome – Purpose, Initial Roadmap
- Paper Session 1: *Governance*
- Paper Session 2: *Safety and Risk*
- Roadmap Review then 10:30 Coffee Break
- Invited Speaker 1
- 12:00 Lunch
- Roadmap Review and Mitigating Risks
- Invited Speaker 2
- Paper Session 3: *Trust Building*
- Invited Speaker 3
- Roadmap Session and 15:30 Coffee Break
- Panel
- Final Roadmap Presentation
- 18:00 Public Panel



Agenda

8am Welcome – Dr. Mark Maybury (Workshop Chair) – A Secure Agent Roadmap (Capabilities, Gaps, Future Research, Outcomes)

8:45 am **Paper Session 1: Governance** [Papers: 15 min presentation, 5 min Q&A]

- Neuro-Symbolic Agents for Safe and Sovereign Industrial Deployments, Daniel Porta and Wolfgang Wahlster. DFKI
- Control Before Consequence: Reframing Infrastructure and Governance for Agentic AI - charlemagnevolt-hub Dr. Bernasco, Aramvolt
- Adaptive Agency: A Human-AI Co-governance Framework for Secure Grassroots Governance in Complex Urban Environments. Xin Ma, Beijing Academy of Social Sciences

9:45 am **Paper Session 2: Safety and Risk** [Papers: 15 min presentation, 5 min Q&A]

- Why Aligned Models Are Not Enough. Amr Gomaa DFKI
- Risk Perspectives in Distributed AI Sysys for Complex Envs. André Meyer-Vitali and Rodica Mihai DFKI

10:25: Quick Draft Roadmap Review - All

10:30-11:00 Coffee Break – Roadmap Discussions

11:30: **Invited Speaker 1:** Open Source Community for Model Security. Stuart Battersby, Red Hat

12:00 LUNCH (Not provided by IJCAI) [Roadmap Discussions Encouraged]

13:00: Quick Roadmap Review - All

13:10 Mitigating Risks Across the Life Cycle of Generative Artificial Intelligence. Mark Maybury. LM Review of Draft Roadmap

13:30pm **Invited Speaker 2:** Nicholas Carlini, Anthropic, formerly Google/DeepMind (Invited)

14:00 **Paper Session 3 Trust Building: Proving AI is Safe and Culturally Appropriate**

- Cryptographic certificates of validity for trustworthy AI. Murdoch J. Gabbay, Heriot-Watt University.
- Designing Identity Before Tools: Human Identity and Trust in Agentic AI. Marco Nieschka & Kordula Hildebrandt, HILDEBRANDT FILM.

14:40 **Invited Speaker 3:** Dr. Florian Tramèr (ETH Zurich). AgentDojo: A Dynamic Environment to Evaluate Prompt Injection. (Invited)

15:15 Final Input to Roadmap

15:30-16:00 Coffee Break – Roadmap Discussions

16:30 Panel: Dr. [Mark Maybury](#), Prof. [Wolfgang Wahlster](#), Ms. [Josephine Liu](#), [Dr. Francesca Rossi](#), Dr. [Mihai Christodorescu](#), Google

17:30 Final Roadmap, Closing Discussion, Future Directions

17:45 End

18:00 Public Panel on Safe Agents and Workshop Results - University Auditorium (capacity 500), the "Keksdose" (cookie tin).

Submission Analysis: Themes and Research Gaps

#	Threat family (why it dominates the next decade)	Key concrete sub-issues	Which of the seven papers raise I
1 Confidentiality	- LLMs are trained on massive personal / proprietary data. Model-inversion, prompt-injection, and data-theft attacks can recover PII, trade-secrets or copyrighted material – producing legal, financial and reputational fallout. - The loss of confidentiality also fuels downstream harms (deep-fakes, credential-harvesting, fraud).	- Model-theft / data-exfiltration - Unauthorized reuse of copyrighted content - Leakage of personal health / biometric data	Paper 1 (Maybury 2026 – §§ 2.2-2.5, 3.1-3.2); Paper 2 (Bernasco 2026 – § 4.3); Paper 3 (Adaptive Agency 2026 – § 1-2); Paper 4 (Gomaa 2026 – §§ 2-3); Paper 5 (Mihai 2026 – §§ 2-3); Paper 6 (Gabbay 2026 – § 1-3).
2 Integrity & Availability (Robustness)	- Poisoned data, jailbreaks, multi-turn persuasive “prompt-injection” attacks, and channel-level attacks (ConVerse benchmark) can force an agent to emit disallowed, false or harmful output. - Hallucination, bias and misinformation remain the most visible failures for regulators, insurers and end-users. - In multi-agent settings the conversation itself is the attack surface (Paper 4).	- Prompt-jailbreak & multi-turn persuasion - Hallucination / bias “Valid-message” attacks that satisfy syntactic checks but are semantically malicious - Distributed-system inconsistencies (different agents hold diverging world models)	Paper 1 (Maybury 2026 – §§ 2.2-2.4, 3.3-3.5); Paper 2 (Bernasco 2026 – § 3 “real-time admissibility”); Paper 3 (Adaptive Agency 2026 – “Dynamic Power Alignment”); Paper 4 (Gomaa 2026 – ConVerse results, dual-firewall architecture); Paper 5 (Mihai 2026 – inconsistent environment models); Paper 6 (Gabbay 2026 – proof-carrying actions that guarantee a policy predicate).
3 Human-Centric / Governance (Identity & Interaction Safety)	- Current AI-safety frameworks (NIST RMF, EU AI Act) focus on code-level alignment, output control and security but ignore the bidirectional impact of sustained human-AI interaction. - The Mirror Hypothesis (Paper 7) shows that AI acts as a cognitive-emotional mirror; the quality of interaction reshapes human identity, trust and decision-making. - Runtime-control (real-time admissibility, bounded execution) and policy-level “Identity-Before-Tools” (Paper 7) are required to keep the human side of the system safe.	- No systematic measurement of interactional effects on cognition/identity - No standards for the emotional/communicative character of agents - No pedagogical or onboarding protocol that guarantees a user’s identity clarity before tool use - Runtime-control gaps (admissibility, bounded execution) that let unsafe actions slip through	Paper 2 (Bernasco 2026 – “Control-Before-Consequence”); Paper 3 (Adaptive Agency 2026 – Dynamic Power Alignment, Mutual Adaptation); Paper 4 (Gomaa 2026 – dual-firewall, compositional safety, policy language); Paper 5 (Mihai 2026 – internal-external risk perspectives, distributed coordination); Paper 6 (Gabbay 2026 – cryptographic certificates of validity, proof-carrying actions); Paper 7 (Identity Before Tools 2024-26 – interactional safety metrics, pedagogical protocols).

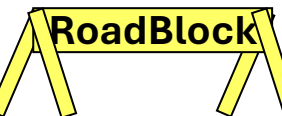
2026-2036 Research Roadmap for Safe Agents

North Star: Ethical, Fair,
Robust, Accountable,
Beneficent

KEY

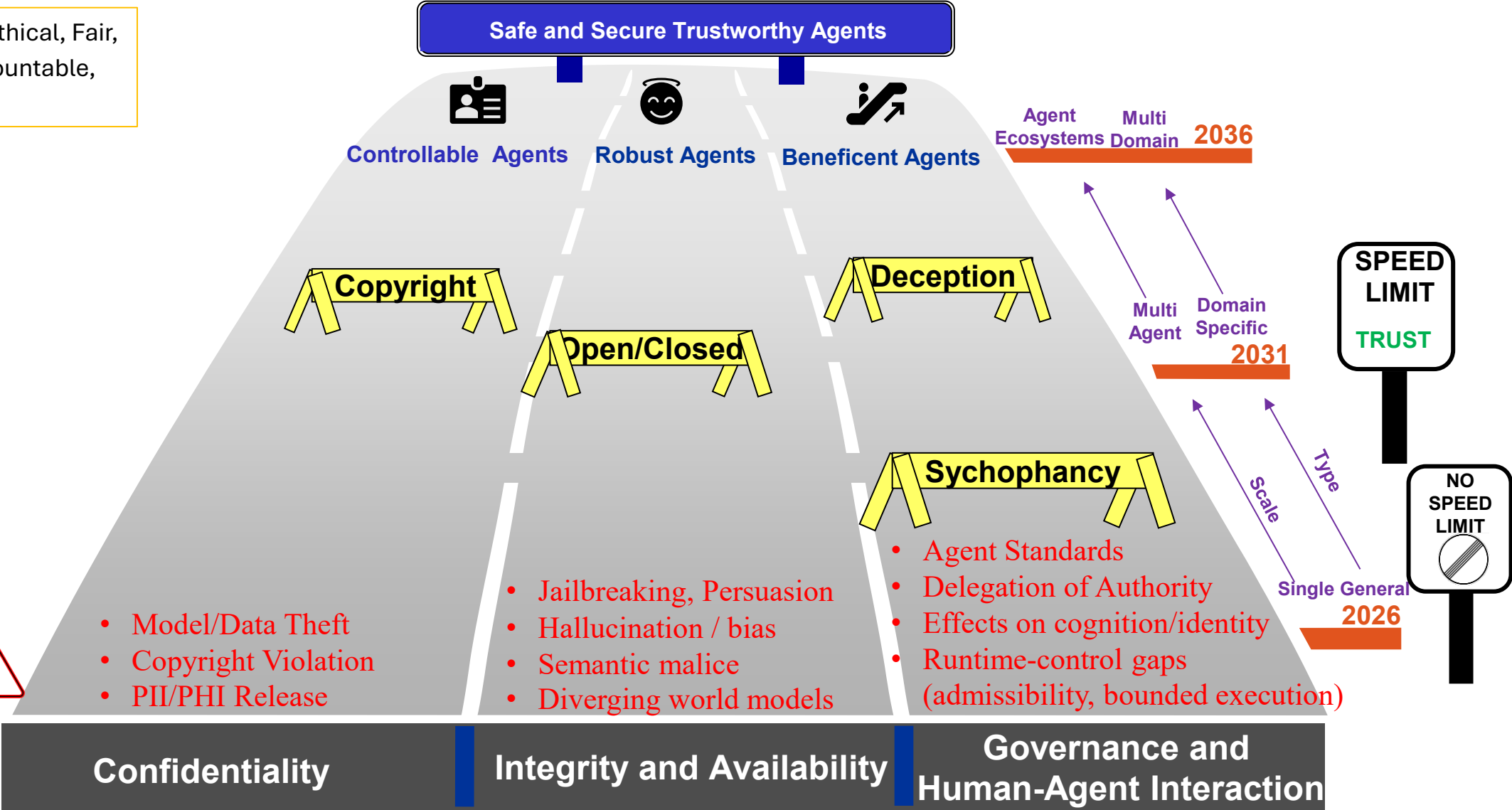
 Threat

 Defense

 RoadBlock

GAPS





2026-2036 Research Roadmap for Safe Agents

Related Fields:
???

Planned
Desired

North Star: Ethical, Fair,
Robust, Accountable,
Beneficent

Safe and Secure Trustworthy Agents

Controllable Agents

Robust Agents

Beneficent Agents

Agent Ecosystems Domain 2036

Differential AI Privacy

Multi-Agent Collaboration

Sustainability

Copyright

Open/Closed

Deception

Explainability

Secure Agent Access

Secure Tool Access

Human-Agent Teaming

Accountability

High Risk (e.g., CBRN) Controls

Alignment

Bounded Execution

Sychophancy

Self-Regulation

Dependency

Milestone

- Model/Data Theft
- Copyright Violation
- PII/PHI Release

Grounding

Guardrails

- Jailbreaking, Persuasion
- Hallucination / bias
- Semantic malice
- Diverging world models

- Agent Standards
- Delegation of Authority
- Effects on cognition/identity
- Runtime-control gaps (admissibility, bounded execution)

Multi Agent Domain Specific 2031

Scale Type Single General 2026

SPEED LIMIT
TRUST

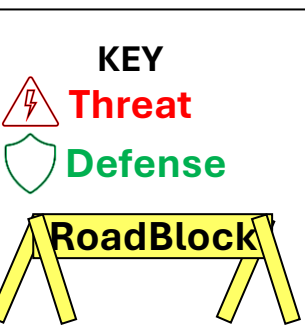
NO SPEED LIMIT

Confidentiality

Integrity and Availability

Governance and Human-Agent Interaction

GAPS



Book Option

Publisher	Key Series / Format	Turnaround Time	Open Access (OA) Options
Springer Nature	LNCS / LNAI, CCIS, SpringerBriefs	Fast (3–5 mos for CCIS/LNCS)	Gold OA & Hybrid options (Book/Chapter level)
MIT Press	AI & Society Series / Standalone Monograph	Standard (9–12 mos)	Select OA models (e.g., Direct to Open)
Cambridge University Press	CUP Computer Science / AI Governance	Standard (8–10 mos)	Gold OA / Read & Publish agreement coverage
CRC Press (T&F)	Security & AI Engineering Series	Fast to Medium (5–7 mos)	Author-funded Gold OA & standard licensing
Wiley / IEEE Press	Wiley-IEEE Press Co-Publications	Medium (6–8 mos)	Hybrid / Gold OA available

Process

- **Book Proposal**
- **Call for Full Paper Submissions**
- **Each submitter Peer review 3 submissions**
- **Introduced by Roadmap Paper**
- **Who is interested in Contributing?**



THANK YOU!
Safe Agents Workshop

THANK YOU

A successful day of learning, and collaboration.
Thank you to all speakers, and organizers!



Safe Agent
Workshop

A Special **THANK YOU**
from the **COMMITTEE**
Safe Agents Workshop



Mark



Wolfgang



Josephine



Francesca



Mihai



LOCKHEED MARTIN 